

MAT282 Intermediate Statistics: Practice Midterm

Franklin University Switzerland · Fall 2026 · Instructor: Assoc. Prof. C. Burke · Practice only — not the real exam

Name: _____ Form: P Score: _____ / 50

Time: about 45 minutes · **Allowed:** pen/pencil, calculator, ruler · **Total:** 50 points · Show working for every calculation: marks are for method as well as answers. Give numerical answers to 2 decimal places unless told otherwise. The data are **synthetic teaching data** from a fictional seaside gelato café, not real records.

ANSWER KEY / MARKING NOTES — instructor copy. Do not distribute.

Scenario. Same café, same 168-day data set, new questions. You work with a **13-day sample**. The manager now wonders whether *humid days are bad for gelato*. **Operationalisation. Humidity** = mean relative humidity (%) during opening hours. **Gelato units** = standard two-scoop cups sold that day. This practice mirrors the real exam's *objectives and timing*, not its tasks or results: do not expect the same pattern of answers.

Part 1 · Concepts (8 points)

1. Hypothesis & falsification [LO 1]

[4]

Three statements are offered: (i) “Humidity affects gelato sales.” (ii) “As humidity rises, daily gelato units fall (negative correlation).” (iii) “Gelato is popular in summer.” For each, say whether a sample could **falsify** it and why (3 × 1 pt). Then rewrite the **weakest** one as a stronger hypothesis (1 pt).

Key: (i) vague, no direction: almost any result is “consistent”; (ii) falsifiable ($r \geq 0$ would count against it); (iii) “popular” and “summer” undefined: not clearly falsifiable. Rewrite earns the 4th point if it names direction, variables and a statistic.

2. Validity & reliability [LO 2]

[4]

For each scenario, say whether the main problem is **validity**, **reliability**, or **neither**, and explain (a, b: 1.5 pts each; c: 1 pt): (a) The humidity sensor reads about 5 points too high on rainy days only. (b) The café uses “units sold” as a measure of customer *satisfaction*. (c) Two staff count cups at closing; their counts differ by a few cups most days.

Key: (a) systematic bias → accuracy/validity of the measurement (accept reliability if explained as inconsistent across conditions); (b) validity: units sold ≠ satisfaction (construct mismatch); (c) reliability (inconsistent repeated measurement).

Part 2 · Describing the data (13 points)

Sample: 13 days.

Day	1	2	3	4	5	6	7	8	9	10	11	12	13
Date	Apr 10	Apr 14	May 24	May 25	Jun 8	Jun 19	Jun 20	Jul 8	Jul 23	Jul 27	Aug 3	Aug 21	Sep 19

Humidity (%)	79	63	80	59	60	75	73	56	61	57	74	65	58
Gelato units	200	189	174	219	223	131	154	201	222	275	205	176	192

3. Centre [LO 3]

[3]

Calculate the mean and median of **humidity**. Mean of gelato units = 197. Do mean and median of humidity agree closely? What does that suggest about the shape?

Key: Humidity mean = $860/13 = 66.15$; median = 63. Close/slightly different: mean > median suggests slight right skew (accept either with reasoning). 1+1+1.

4. Quartiles & fences [LO 3]

[4]

Find Q1 and Q3 of humidity (Sheets convention: 4th and 10th of 13 ordered values), the IQR and the $1.5 \times \text{IQR}$ fences. Is there an outlier? **Prove** it with your fences.

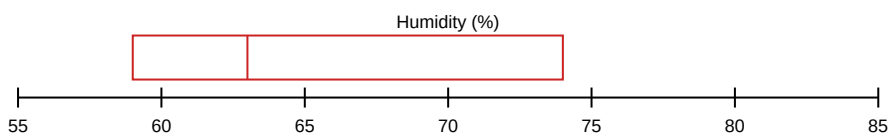
Key: Sorted: [56, 57, 58, 59, 60, 61, 63, 65, 73, 74, 75, 79, 80]. Q1=59, Q3=74, IQR=15, fences 36.5–96.5. Min 56, max 80 → no outlier. (1+2+1)

5. Box plot [LO 4]

[3]

Draw a **box plot** of humidity on the number line below (box at Q1–Q3, median line, whiskers to the smallest and largest non-outlying values).

Key: Box 59–74, median line 63, whiskers 56 and 80. 1 box, 1 median, 1 whiskers.



6. Standardising [LO 3, 5]

[3]

Humidity: mean 66.2, sd 8.8. Gelato units: mean 197.0, sd 35.6. On **Apr 14** humidity was 63 % and 189 units were sold. Calculate a **z-score** for each and say which was more unusual for the café.

Key: $z(\text{hum}) = (63 - 66.2)/8.8 = -0.36$; $z(\text{units}) = (189 - 197.0)/35.6 = -0.22$. More unusual = larger $|z|$: humidity. 1+1+1.

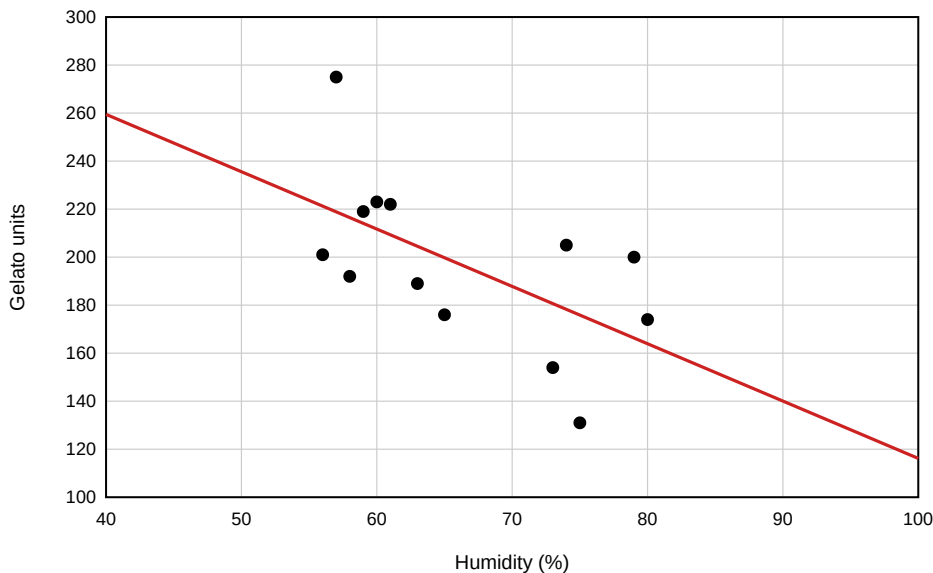
Part 3 · Relationship and prediction (13 points)

7. Scatterplot [LO 4]

[3]

Plot humidity (x) against gelato units (y) and draw a line of best fit by eye.

Key: Negative slope. Computed line $\hat{y} = 354.99 - 2.39x$. 2 pts points, 1 pt line.



8. Reading correlation [LO 3, 6]

[2]

Four scatterplots (A–D) have $r = -0.9$, -0.5 , 0 and $+0.7$ in some order. Describe in words how each looks, then say which r you would assign to **your plot** and why. (Imagine: A tight upward cloud; B a round blob; C a very tight downward line; D a loose downward cloud.)

Key: A=+0.7, B=0, C=-0.9, D=-0.5. Computed r for sample = -0.59 → closest to D. 1+1.

9. Prediction & error [LO 5]

[4]

A fitted line is $\hat{y} = 354.99 - 2.39 \cdot \text{humidity}$. (a) Predict gelato units for humidity 50 % and 70 %. (2) (b) On Jun 8 (humidity 60, actual 223) and Jul 27 (humidity 57, actual 275), compute each **residual** and the **mean absolute deviation**. (2)

Key: (a) 235.5 and 187.7. (b) residual 11.4, residual 56.2; MAE = 33.8.

10. Variable weights [LO 5]

[2]

To forecast **next Saturday's** gelato units the manager weights the last three Saturdays: 50 % on the latest (232 units), 30 % on the one before (205) and 20 % on the oldest (190). Calculate the weighted forecast. Would the plain average be higher or lower, and why?

Key: $0.5 \cdot 232 + 0.3 \cdot 205 + 0.2 \cdot 190 = 215.5$; plain mean = 209.00; weighted is higher because recent (larger) values carry more weight. 1+1.

11. Repair the conclusion [LO 6, 7]

[2]

A student writes: “ $r = -0.59$, so humidity explains 59% of the variation in gelato units and humidity causes lower sales.” Identify **two errors** and correct them.

Key: (1) r is not % explained: $R^2 = r^2 = 0.35 \rightarrow$ about 35%. (2) correlation (observational) does not establish cause: temperature, events, weekends may drive both.

Part 4 · Comparing periods and significance (6 points)

12. Difference of means [LO 3, 6, 7]

[6]

On 30 June the café **cut** the iced coffee price from €3.50 to €3.10. Daily iced coffee units — before: $n=85$, mean 101.4, sd 18.0; after: $n=83$, mean 124.5, sd 17.2. (a) Calculate t for the difference in means (cutoff $|t| > 1.97$). (3) (b) A colleague says “the huge t proves the price cut worked.” Give **two** reasons that is too strong. (2) (c) Name one extra variable you could add to a LINEST model to adjust for a rival explanation. (1)

Key: (a) $SE=2.71$, $diff=23.08$, $t=8.51 \rightarrow$ significant (1+1+1). (b) any two: observational, season/temperature confound, significance $\neq$ importance, “proves” overstated. (c) temperature, beach_event, weekend, sunshine.

Part 5 · Computation: LINEST and Sheets (6 points)

13. LINEST [LO 5, 6]**[6]**

LINEST for **iced_coffee_units** on **sunshine_hours**, **post_change**, **weekend** (in that order), TRUE, TRUE:

	Col 1	Col 2	Col 3	Col 4
Row 1	17.42	16.96	3.23	76.28
Row 2	2.39	2.33	0.47	3.40
Row 3	0.56	14.01	#N/A	#N/A
Row 4	70.55	164.00	#N/A	#N/A
Row 5	41556.68	32201.03	#N/A	#N/A

Row 1 coefficients are reversed, intercept last.

(a) Use the coefficients to predict iced coffee units for a **weekend** day with **8 hours of sunshine** after the policy change (post_change = 1). (2)

(b) Compute the t-statistic for sunshine_hours and for post_change. Which predictor is statistically stronger, and what does this suggest about the pre/post difference in Question 12? (2)

(c) A classmate types =LINEST (C2:C169, A2:C168, TRUE, TRUE) and gets #REF!/#VALUE!. Spot the error. (2)

Key: (a) coefficients: weekend 17.42, post 16.96, sun 3.23, intercept 76.28 → 136.5 (1 setup, 1 value). (b) $t(\text{sun})=6.85$, $t(\text{post})=7.27$; post_change has the larger $|t|$; both are significant, so the post-policy rise in coffee sales remains after controlling for sunshine and weekend, though a price cut is still not proven as the cause (1+1). (c) the y range has 168 rows (C2:C169) but the X range has 167 (A2:C168): ranges must be the same length. (2)

Part 6 · Free thinking (4 points)

14. Explain the surprise [LO 1, 7]

[4]

In the full data the correlation between **temperature and pastry units** is $r = -0.26$ (weak, negative). The owner is surprised, since “good weather is good for business.” In 4–5 sentences give **two different explanations** (one about the *data/measurement*, one about the *real world*) and say **which you would test first**, how, and what result would change your mind.

Key: Rubric: data explanation (1) e.g. weak/noisy measure, sampling, outliers; real-world mechanism (1) e.g. people shift from pastries to cold items in heat; a feasible test in Sheets (1) e.g. CORREL by month, scatter, add to LINEST; falsification/“what would change my mind” (1).